

Practice Quiz: Perception / Bayesian Mechanics

(Advanced Theory)

Name: __ Date: __

Part A: Multiple Choice

1. The particular partition divides system states into: A) Two groups B) Four groups: external (η — hidden environment), sensory (s — input), active (a — output), and internal (μ — beliefs), where $s + a$ form the Markov blanket C) Random groups D) Three groups
2. The Free Energy Lemma states that: A) All systems minimize energy B) Under NESS with a Markov blanket, internal state dynamics approximate gradient descent on variational free energy C) Free energy is always zero D) Physics is irrelevant
3. The Helmholtz decomposition splits dynamics into: A) Fast and slow B) Dissipative (gradient flow — inference) and solenoidal (rotational — exploration) components C) Linear and nonlinear D) Deterministic and stochastic only
4. Solenoidal flow is essential because: A) It dissipates energy B) It allows complex, oscillatory dynamics while preserving the steady-state density — without it, all behavior would be simple gradient descent C) It causes noise D) It is always zero
5. The path-integral formulation connects Active Inference to: A) Computer science only B) The principle of least action in physics — the most likely trajectory minimizes an action functional related to free energy C) Random walks D) Digital computation
6. Dual-aspect monism means: A) There are two separate substances B) Physical dynamics and inferential dynamics are mathematically equivalent descriptions — neither is more fundamental C) Only physics matters D) Only inference matters
7. Gauge symmetry in Bayesian mechanics reflects: A) Physical symmetry only B) The non-unique mapping from internal states to beliefs — different neural configurations can encode the same belief C) Measurement error D) Randomness

Part B: Essay Questions

1. Derive the Free Energy Lemma from the particular partition. Starting from the Markov blanket condition and NESS assumptions, show how internal state dynamics can be expressed as free energy minimization. Identify each mathematical step and the assumptions required. At which step could the argument fail for biological systems? (500 words)
2. Is the Free Energy Principle unfalsifiable? This is the most common philosophical critique. Address it by: (a) distinguishing between the FEP as a mathematical framework vs. as an empirical claim, (b) identifying what would need to be shown to refute the FEP, (c) comparing to other foundational principles (conservation of energy, natural selection) that have been called tautological. (400 words)
3. How does Bayesian mechanics relate to thermodynamics? Both involve free energies, both involve entropy, both describe steady-state systems. Are these deep connections or superficial analogies? Discuss the Helmholtz free energy in thermodynamics vs. variational free energy in Active Inference. What does the connection to the second law of thermodynamics reveal? (400 words)